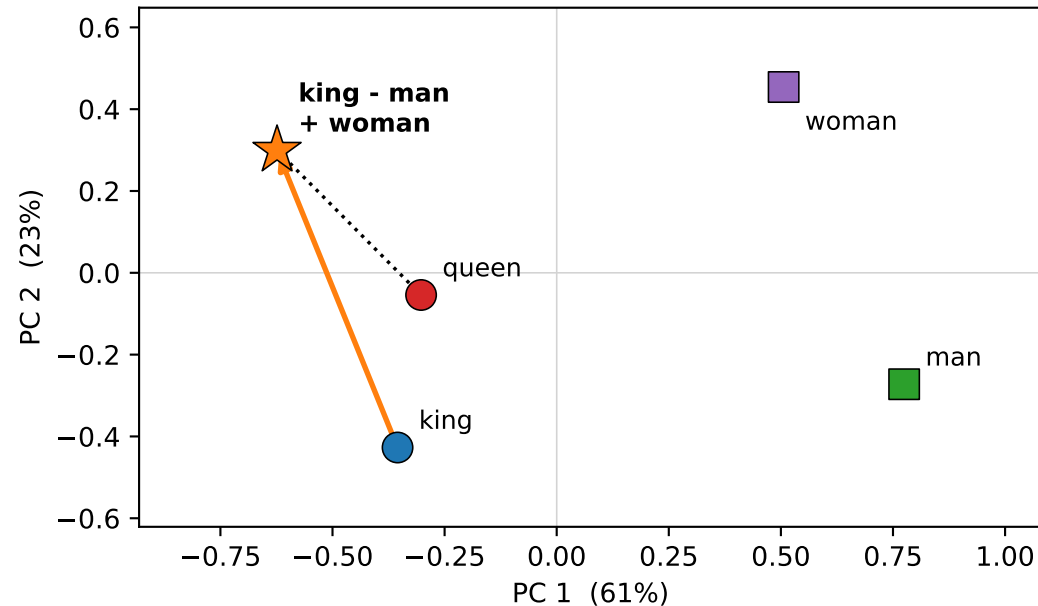


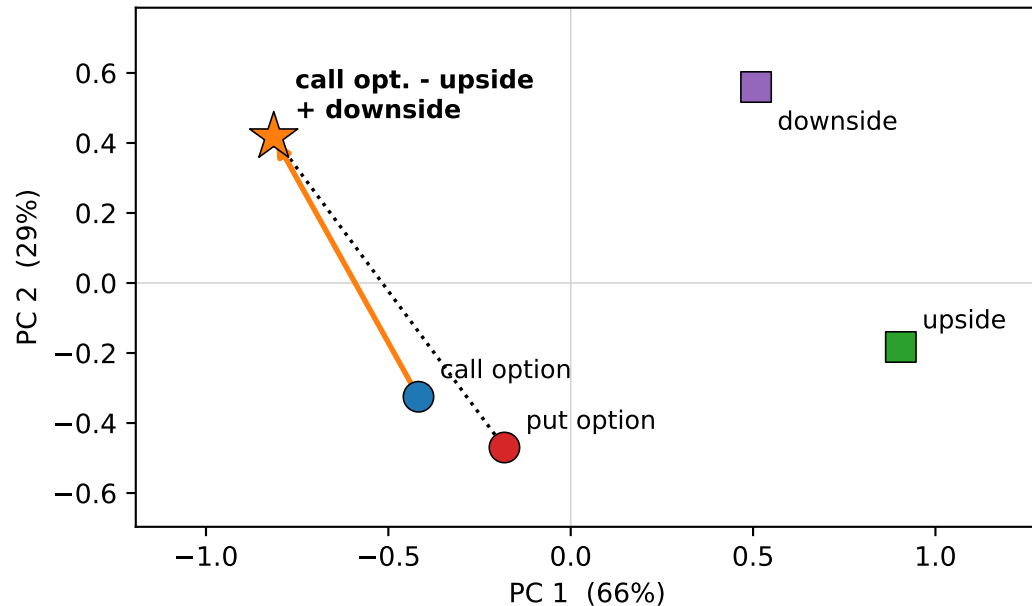
Word-vector arithmetic in pre-trained GloVe (300d), projected to 2D by PCA

(a) Classic analogy holds: nearest word is *queen*
 $king - man + woman \approx queen$ ($\cos = 0.67$)



nearest words to result:
queen 0.67
princess 0.54
throne 0.54
parallelogram $\cos = +0.40$

(b) Finance analogy fails: nearest words are generic
 $call - upside + downside$ does not give *put* in GloVe



nearest words to result:
consider 0.53
options 0.50
choice 0.49
'put option' = 0.56 (shortlist only)
parallelogram $\cos = -0.15$